

Table 1. English TruthfulQA results with evaluated-model KV cache disabled. Repeated rows report mean $\pm$ standard error; compact single-pass rows report means without an error term. TBD marks unrun model-method pairs.

	Method	True (%) $\uparrow$	Informative (%) $\uparrow$	Truthful-concept relevance (0–2) $\uparrow$	Instruction relevance (0–2) $\uparrow$	Fluency (0–2) $\uparrow$
Gemma-2-2B	Original	49.79 $\pm$ 0.22	95.35 $\pm$ 0.22	TBD	1.35 $\pm$ 0.02	1.23 $\pm$ 0.01
	ITI	50.65 $\pm$ 0.39	95.01 $\pm$ 0.13	TBD	1.39 $\pm$ 0.02	1.29 $\pm$ 0.01
	ActAdd	60.37 $\pm$ 0.66	86.49 $\pm$ 0.61	TBD	1.22 $\pm$ 0.01	1.08 $\pm$ 0.01
	Mean-AcT	48.00 $\pm$ 0.51	93.76 $\pm$ 0.34	TBD	1.39 $\pm$ 0.02	<u>1.29 $\pm$ 0.01</u>
	Linear-AcT	46.71 $\pm$ 0.76	94.15 $\pm$ 0.12	TBD	<u>1.41 $\pm$ 0.01</u>	1.28 $\pm$ 0.01
	PID-AcT	47.07 $\pm$ 0.52	93.83 $\pm$ 0.34	TBD	1.38 $\pm$ 0.01	1.29 $\pm$ 0.01
	ODESteer	56.18 $\pm$ 0.29	<u>95.76 $\pm$ 0.26</u>	TBD	1.37 $\pm$ 0.02	1.23 $\pm$ 0.01
	S-PID	52.51 $\pm$ 0.44	94.59 $\pm$ 0.30	TBD	1.39 $\pm$ 0.01	1.22 $\pm$ 0.00
	A-LQR	62.33 $\pm$ 0.25	91.65 $\pm$ 0.39	TBD	1.29 $\pm$ 0.02	1.15 $\pm$ 0.01
	H_∞ (ours)	70.11 $\pm$ 0.70	79.34 $\pm$ 0.50	TBD	1.14 $\pm$ 0.01	1.07 $\pm$ 0.01
Llama-3-8B	Original	49.28 $\pm$ 0.98	96.99 $\pm$ 0.29	TBD	1.33 $\pm$ 0.03	1.30 $\pm$ 0.01
	ITI	57.14 $\pm$ 0.79	97.53 $\pm$ 0.09	TBD	1.38 $\pm$ 0.02	1.30 $\pm$ 0.01
	ActAdd	56.72 $\pm$ 0.99	88.69 $\pm$ 0.82	TBD	1.23 $\pm$ 0.02	1.17 $\pm$ 0.01
	Mean-AcT	47.47 $\pm$ 0.71	97.55 $\pm$ 0.26	TBD	1.36 $\pm$ 0.01	1.31 $\pm$ 0.01
	Linear-AcT	48.20 $\pm$ 0.69	97.50 $\pm$ 0.19	TBD	1.35 $\pm$ 0.02	<u>1.33 $\pm$ 0.00</u>
	PID-AcT	47.69 $\pm$ 0.21	97.60 $\pm$ 0.26	TBD	1.34 $\pm$ 0.01	1.31 $\pm$ 0.01
	ODESteer	<u>95.81 $\pm$ 0.06</u>	9.28 $\pm$ 0.36	TBD	0.01 $\pm$ 0.00	0.03 $\pm$ 0.01
	S-PID	57.55 $\pm$ 0.21	97.80 $\pm$ 0.27	TBD	1.41 $\pm$ 0.02	1.27 $\pm$ 0.01
	A-LQR	58.95 $\pm$ 0.27	<u>98.07 $\pm$ 0.20</u>	TBD	<u>1.44 $\pm$ 0.03</u>	1.21 $\pm$ 0.01
	H_∞ (ours)	89.30 $\pm$ 0.47	60.81 $\pm$ 0.34	TBD	0.69 $\pm$ 0.02	0.91 $\pm$ 0.01
Qwen-2.5-14B	Original	55.89 $\pm$ 0.33	96.47 $\pm$ 0.15	1.38 $\pm$ 0.01	1.46 $\pm$ 0.01	1.24 $\pm$ 0.01
	ITI	61.62 $\pm$ 0.62	95.32 $\pm$ 0.35	1.45 $\pm$ 0.02	1.52 $\pm$ 0.01	1.25 $\pm$ 0.01
	ActAdd	64.04 $\pm$ 0.42	87.61 $\pm$ 0.61	1.08 $\pm$ 0.01	1.17 $\pm$ 0.01	0.99 $\pm$ 0.01
	Mean-AcT	55.35 $\pm$ 1.09	96.03 $\pm$ 0.21	1.39 $\pm$ 0.01	1.50 $\pm$ 0.01	1.24 $\pm$ 0.01
	Linear-AcT	56.28 $\pm$ 0.95	96.18 $\pm$ 0.28	1.38 $\pm$ 0.01	1.50 $\pm$ 0.01	1.23 $\pm$ 0.01
	PID-AcT	56.08 $\pm$ 0.89	96.28 $\pm$ 0.15	1.39 $\pm$ 0.02	1.49 $\pm$ 0.01	1.24 $\pm$ 0.01
	ODESteer	61.40 $\pm$ 0.60	94.00 $\pm$ 0.15	1.34 $\pm$ 0.01	1.41 $\pm$ 0.01	1.21 $\pm$ 0.00
	S-PID	66.93 $\pm$ 0.93	96.52 $\pm$ 0.22	1.47 $\pm$ 0.00	1.51 $\pm$ 0.01	<u>1.26 $\pm$ 0.01</u>
	A-LQR	71.82 $\pm$ 0.60	<u>96.99 $\pm$ 0.28</u>	<u>1.52 $\pm$ 0.01</u>	<u>1.53 $\pm$ 0.01</u>	<u>1.26 $\pm$ 0.01</u>
	H_∞ (ours)	85.31 $\pm$ 0.33	89.40 $\pm$ 0.35	1.39 $\pm$ 0.01	1.34 $\pm$ 0.01	1.15 $\pm$ 0.00
Qwen-2.5-32B	Original	61.86	98.53	1.37	1.49	1.27
	ITI	TBD	TBD	TBD	TBD	TBD
	ActAdd	TBD	TBD	TBD	TBD	TBD
	Mean-AcT	TBD	TBD	TBD	TBD	TBD
	Linear-AcT	TBD	TBD	TBD	TBD	TBD
	PID-AcT	TBD	TBD	TBD	TBD	TBD
	ODESteer	TBD	TBD	TBD	TBD	TBD
	S-PID	73.84	96.82	1.47	1.40	<u>1.26</u>
	A-LQR	78.73	<u>97.31</u>	<u>1.51</u>	<u>1.53</u>	1.23
	H_∞ (ours)	82.15	92.42	1.43	1.42	1.16